

SEISMIC ACQUISITION PARAMETERS

The 3-D source/receiver geometry consisted of east-west receiver lines spaced 1,320 ft apart and north-south source lines spaced 880 ft apart. Linear, 12-element receiver arrays were constructed so each array spanned 110 ft. The distance between the centers of adjacent arrays was 110 ft, which means there was a continuous ground coverage of uniformly spaced geophones along each receiver line (fig. 7). Vibrator points (VP's) were spaced at intervals of 220 ft. Six sweeps from four inline vibrators were summed at each VP. A linear, 8–120 Hz sweep rate was used, and the sweep length was 14 s. The four vibrators were stationed bumper to bumper with their pads separated 35 ft to create a source array 135 ft long. To minimize ground damage, the vibrators moved forward one pad width after each sweep. At each VP, the final positions of the 24 pad imprints created by the four-element, six-sweep summation were symmetrically distributed on each side of the source flag (fig. 7).

BINNING

The receiver spacing of 110 ft and the source spacing of 220 ft created an acquisition bin measuring 55 by 110 ft. After the 3-D data volume was stacked, the data were interpolated in the source line direction to create interpretation bins measuring 55 by 55 ft.

DATA PROCESSING

The major data processing steps were zero-phase correction of field records, prestack Q compensation to emphasize higher frequencies, surface consistent deconvolution and statics, iterative velocity analysis, stacking, and one-pass 3-D wave equation migration.

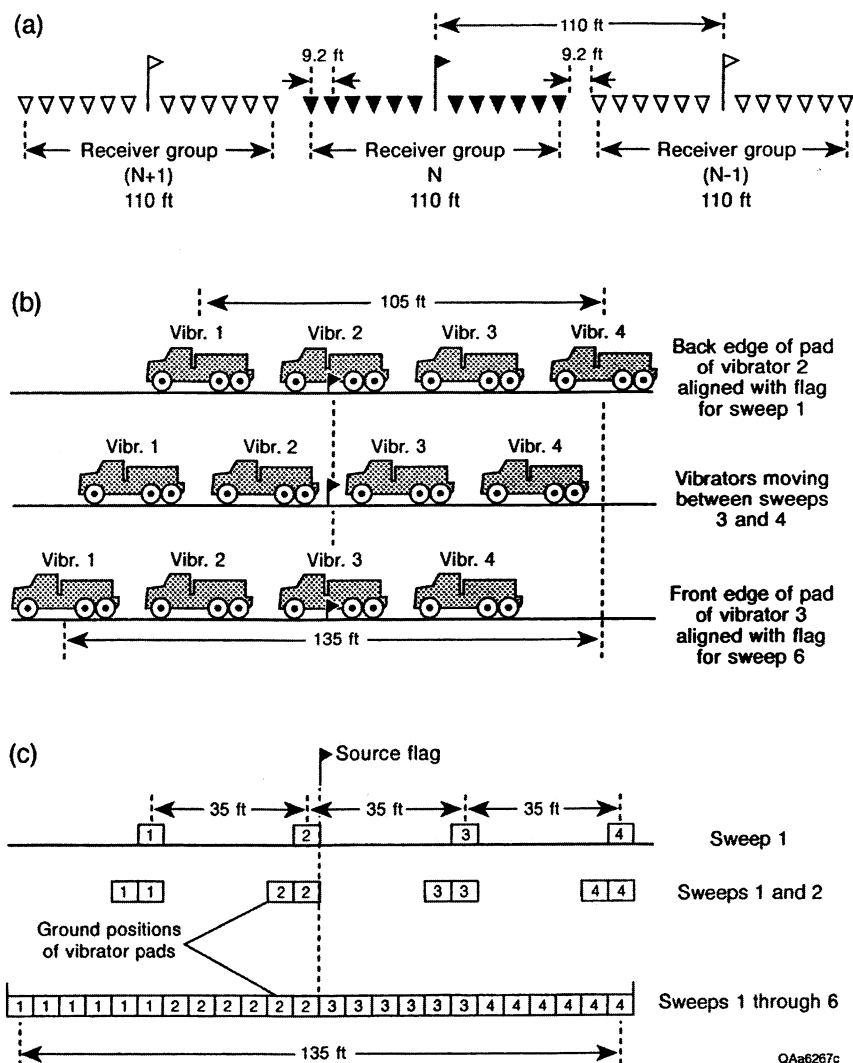


Figure 7. (a) Map of receiver-array geometry used to create uniform ground coverage along each receiver line. (b) Side view of the position and movement of the four-vibrator source array relative to the source flag as six sweeps were recorded to create the field record at each source point. Each rectangular box labeled "Vibr. 1," and so forth, represents the vibrator vehicle. The vibrator pad is located near the center of each box. Vibrator movement between sweeps is from right to left. (c) Map view showing the pad positions of vibrators 1 through 4 relative to the source flag as six sweeps were recorded.